

Learning Dynamics in Anxiety: The Role of Punishment Sensitivity and Learning Rate in Sequential Evaluation

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Abstract

Anxiety is fundamentally an anticipatory response to uncertain future threats, making sequential evaluation crucial. However, most computational research has involved one-step tasks that identify an elevated punishment learning rate as a key feature of anxious behavior, while some studies also suggest a lack of evidence for the role of the punishment sensitivity parameter. This contradicts recent research involving sequential evaluation tasks, which supports the idea that anxiety is primarily an uncertainty disorder. Nevertheless, these experiments have mainly focused on model-based algorithms and have left the role of the punishment learning rate unexplored. To reconcile the differences in the literature and better understand how these two parameters influence anxiety, two hybrid models were developed: one with differentiated learning rates for rewards and punishments, and another with differentiated sensitivity parameters. The models were then evaluated in the Cliff Walking task, using a deterministic environment and incorporating stochasticity in action selection to simulate that an agent lacks complete control over its decisions. The results suggest that the impact of estimated punishments on planning is more significant than the speed at which they are learned, highlighting the importance of heightened punishment sensitivity in anxiety-related behaviors such as avoidance, risk aversion, threat overestimation, and fear generalization.

Keywords: anxiety; reinforcement learning; computational psychiatry; sequential evaluation

Introduction

Computational modeling for studying anxiety is of great importance as anxiety disorders have the highest incidence worldwide among mental disorders (Stein, Scott, de Jonge, & Kessler, 2017). Specifically, the reinforcement learning (RL) approach has aimed to construct mechanistic models that explain the underlying components and their interactions involved in anxious decision-making (Raymond, Steele, & Seriès, 2017; Pike & Robinson, 2022), where an altered part of the model describes anxious behavior (Huys, 2014), for example, a higher value in a certain parameter. However, there are mixed results regarding whether a higher punishment learning rate or an enhanced punishment sensitivity parameter plays a key role in anxious decision-making. This discrepancy becomes even more relevant when considering that these different conclusions stem from distinct kinds of tasks and, consequently, different types of models.

On one hand, most anxiety computational research has focused on simplified one-step tasks, such as multi-armed bandits, where an agent performs many trials but each trial ends after a single action, meaning the agent only needs to

learn the appropriate response for a single environmental state (Yamamori & Robinson, 2023). Additionally, a systematic review and meta-analysis of 27 articles by Pike and Robinson (2022) supports that, in studies using one-step tasks, an enhanced punishment learning rate is the primary factor for modeling anxious decision-making, while also indicating a lack of evidence for the role of punishment sensitivity. Still, due to the nature of these tasks, planning has not been considered and the models developed only try to replicate direct learning in one-state environments. Nevertheless, anxiety is an anticipatory response to uncertain future threats perceived as uncontrollable and highly aversive (Clark & Beck, 2010), involving a multi-step process in which it should be considered how planning may be biased (Sharp, 2025).

On the other hand, anxiety computational research involving sequential evaluation tasks is more recent and as a result comprises only a few studies, which primarily rely on variations of the value iteration algorithm to model biased offline planning and reproduce common behaviors observed in many anxiety disorders, such as avoidance, risk aversion, threat overestimation, and fear generalization (Zorowitz, Momennejad, & Daw, 2020; Gagne & Dayan, 2022). These works have found that heightened punishment sensitivity is a key feature to reproduce anxious behavior, highlighting that anxiety is mainly a disorder of uncertainty (Brown, Price, & Dombrowski, 2023), although the role of the punishment learning rate remains unexplored in multi-step tasks.

Therefore, while earlier studies on modeling anxious behavior mainly emphasize the importance of how quickly punishment information is integrated over time, what matters in sequential evaluation experiments is how much an agent anticipates and dislikes being punished. As computational research on anxiety is relatively recent, no study has tested the contributions of both parameters—punishment learning rate and punishment sensitivity—simultaneously in a sequential evaluation task, which is a more natural setting for anxiety. To address this, it is necessary to use hybrid models that retain the core components of both approaches: direct learning in the environment and offline planning. Ultimately, this will help reconcile differences in the literature and offer deeper insights into the learning dynamics of anxiety in sequential evaluation tasks.

Among hybrid RL models, empirical evidence suggests that a hybrid successor representation (SR) is the most effective

tive algorithm for replicating human decision-making in sequential evaluation tasks (Momennejad et al., 2017; Russek, Momennejad, Botvinick, Gershman, & Daw, 2017). Specifically, SR-Dyna is considered the best model for capturing human retrospective revaluation behavior (Momennejad, 2020). Moreover, online planning is employed in the basic SR algorithm when successor representations are estimated or used to compute the estimation of another variable (Gershman, 2018). Therefore, the SR-Dyna model implements direct learning in the environment, online planning, and offline planning. The integration of these three mechanisms—with some modifications—may better reflect how individuals with anxiety process information in real-world situations, compared to models that incorporate only one or two of them. For example, for a woman with acrophobia, direct learning instantly increases her association of heights with threat after falling from a cliff. Meanwhile, offline planning involves rumination on various possible situations based on past experiences while at rest. Finally, online planning manifests as hypervigilance for potential future threats when she’s actually at heights and has to consider path estimations, with all three mechanisms reinforcing her fear of heights.

Furthermore, in SR family models, rewards are learned independently from successor representations (trajectories), but both can later be used to compute Q-values for each state in an environment (Ducarouge & Sigaud, 2017). In particular, this independence in learning allows for specific modifications to the model’s equations to assess the impact of differentiated learning rates and sensitivity parameters, taking into account distinct learning dynamics. For example, a first model can be constructed by implementing differentiated learning rates in the reward function. In contrast, a second model can be designed with differentiated sensitivity parameters for rewards and punishments in the successor function. Next, simulations using different values of these parameters can assess the impact on anxious behavior of a higher punishment learning rate in the first model and an enhanced punishment sensitivity parameter in the second one. Hence, we propose using an SR-Dyna algorithm as a basis to design new models to address the objective of this work.

Methodology

The present work adopts a theoretical modeling approach to generate predictive insights for anxiety research. Two models were developed to simulate agents with varying levels of punishment learning rate or punishment sensitivity, with the aim of evaluating their respective contributions to anxious decision-making within a Markov Decision Process (MDP).

Models

In algorithm 1, we present the pseudocode for an SR-Dyna model considering a deterministic environment, incorporating temporal difference (TD) learning and estimating Q-values, followed by a detailed explanation of the modifications introduced to it for the first and second models proposed in this work.

Algorithm 1 : SR-Dyna

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1: Initialize  $R(s) \forall s \in S$ .
2: Initialize  $M(s, s', a) \forall s, s' \in S$  and  $a \in A(s)$ .
3: Initialize  $Q(s, a) \forall s \in S$  and  $a \in A(s)$ .
4: Initialize  $E(s, a) \forall s \in S$  and  $a \in A(s)$ .
5: while episode  $\leq N$  do
6:    $s \leftarrow$  actual state
7:    $a \leftarrow \epsilon$ -greedy( $s, Q$ )
8:   Execute action  $a$ ; observe the received reward ( $r$ ), and the
   next state ( $s'$ ).
9:    $a^+ \leftarrow \arg \max_a Q(s', a)$ 
10:   $M(s, s', a) \leftarrow M(s, s', a) + \alpha(I_{[s=s']} +$ 
    $\gamma M(s', s'', a^+) - M(s, s', a))$ 
11:   $R(s) \leftarrow R(s) + \alpha(r - R(s))$ 
12:   $Q(s, a) \leftarrow \sum_{s'} M(s, s', a) R(s')$ 
13:   $E(s, a) \leftarrow r, s'$ 
14:  for  $n$  do
15:     $s \leftarrow$  previously observed random state.
16:     $a \leftarrow$  random action previously executed in  $s$ .
17:     $r, s' \leftarrow E(s, a)$ 
18:    Execute steps from 9 to 12 using the new values for  $s$ ,
     $a$ ,  $r$  and  $s'$ .
19:  end for
20: end while
Parameters:  $N, \epsilon, \alpha, \gamma$  and  $n$ .

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First, in Algorithm 1 the vector for rewards (R), the successor tensor (M), the Q-value matrix (Q) and the matrix with the environment’s model (E) are initialized with zeros in each entry. Then, during each episode in the model-free part of the model, the agent selects an action (a) using an ϵ -greedy policy, executes it, receives a reward (r) and observes the next state (s'). Later, the agent updates the current successor representation using the TD method, incorporating both immediate and discounted future occupancy predictions. Additionally, the estimated reward, Q-value estimates, and the environment’s model are updated. In the Dyna-style offline planning phase (steps 14-18), the algorithm replays stored experiences using the environment’s model and updates the SR model estimations.

Model 1: Dyna α -SR This model differs from Algorithm 1 solely in how the learning rate (α) is selected in the reward function (step 11). Specifically, the α value now depends on whether the agent receives a reward or a punishment, as shown in Equation 1, in order to have differentiated learning rates.

$$\alpha = \begin{cases} \alpha^+ & \text{if } r \geq 0 \\ \alpha^- & \text{if } r < 0 \end{cases} \quad (1)$$

Model 2: Dyna β -SR Building on Algorithm 1 and the work of Zorowitz et al. (2020) to model anxious behavior, we introduce a novel variant: Dyna β -SR. In this new model, it is necessary not only to calculate the action with the highest value (a^+) for the next state, but also the lowest (a^-), as

shown in Equations 2 and 3.

$$a^+ \leftarrow \arg \max_a Q(s', a') \quad (2)$$

$$a^- \leftarrow \arg \min_a Q(s', a') \quad (3)$$

Later, a^+ and a^- are used in the β^M module (Eq. 4) to estimate the successor representations with the highest and lowest action for the next state, which are multiplied by ω and its complement, respectively. Therefore, we can say that a pessimistic agent ($\omega = 0$) is sensitive to the paths that lead to punishments or threats, considering that it lacks complete control over its actions. This can be considered as a low level of self-efficacy, reflecting a lack of belief in their ability to perform the necessary behaviors to achieve specific performance goals (Bandura, 1978; Zorowitz et al., 2020). In contrast, an optimistic agent ($\omega = 1$) anticipates that its future actions will fully align with its preferences to maximize rewards (Zorowitz et al., 2020), being sensitive to the paths that lead to them. Finally, β^M is used in the successor function (Eq. 5) as the successor representation for the next state.

$$\beta^M \leftarrow \omega M(s', s'', a^+) + (1 - \omega) M(s', s'', a^-) \quad (4)$$

$$M(s, s', a) \leftarrow M(s, s', a) + \alpha (I_{[s=s']} + \gamma \beta^M - M(s, s', a)) \quad (5)$$

Experimental Task

A variation of the Cliff Walking task (Fig. 1) was implemented in a 9x9 gridworld with a discrete-time environment, finite horizon, and deterministic dynamics. The stochasticity lies in the agent’s action selection, as it follows an ϵ -greedy policy to choose the action it executes in each state. This policy is used both during the agent’s training phase ($N = 200$ episodes) and in the final test (1 rollout) to simulate that the agent does not have complete control over its actions, as is the case in many real-world scenarios.

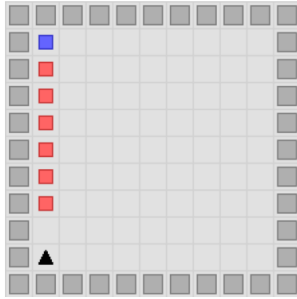


Figure 1: **Virtual environment of the Cliff-Walking task.** The triangle indicates the agent’s starting position, the blue square marks the goal, the red squares represent aversive states (the cliff), and the light gray squares are neutral, where neither a reward nor a punishment is received.

The agent’s objective in this environment is to reach the goal without falling into an aversive state. At the beginning, the agent is unfamiliar with the environment and must explore it to find the optimal route. The agent can perform four possible actions: moving up, down, right, or left, as long as it does not collide with a wall (dark gray squares). Finally, an episode ends under the following conditions: the agent reaches the goal (reward = 1), falls into an aversive state (punishment = -1), or the maximum step limit per episode is reached (100).

Additionally, the anxious behaviors we aimed to evaluate in the experimental task were operationalized as follows:

- **Threat overestimation:** states that would otherwise be neutral are estimated as signaling threat unrealistically.
- **Fear generalization:** neutral states distant from aversive ones are considered dangerous.
- **Avoidance:** the agent consistently moves away from aversive states.
- **Risk aversion:** instead of exploring more of the environment, the agent exploits a safe zone of the state space.

Simulations

Three simulations were conducted for the Dyna α -SR agents with α^- values of 0.05, 0.075, and 0.1, respectively, while keeping $\alpha^+ = 0.05$ constant in all cases. In addition, we ran three simulations for the Dyna β -SR agents with ω values of 1.0, 0.5, and 0.0, respectively. The parameters shared between the models were set as follows: $\alpha = 0.05$ (for the successor function), $\gamma = 0.9$, $\epsilon = 0.2$ and $n = 3$ (recall). All simulations were performed using the NeuroNav library (Juliani, Barnett, Davis, Sereno, & Momennejad, 2022) as a foundation for programming the agents and the experimental task. The complete code is publicly available at the following link: <https://github.com/Alicia-MJ/RL-Anxiety-MathPsy-ICCC.git>

Results

For each agent, a heatmap of the environment was generated based on the final estimated value of each state, computed by averaging its corresponding Q-values. In these maps, values close to 1.0 (the maximum possible value) appear in deep blue, values near 0 are shown in white, and values close to -1.0 (the minimum possible value) are displayed in dark red. Arrows indicate the highest-value action estimated for each state, while the rollout corresponding to the evaluation phase is highlighted with black arrows. Additionally, as a support to the main analysis, the total number of steps and the total return received per episode were graphed for each agent to evaluate its evolution across episodes. Together, these three analyses provide a more comprehensive assessment of each agent’s behavior and learning.

Dyna α -SR Agents

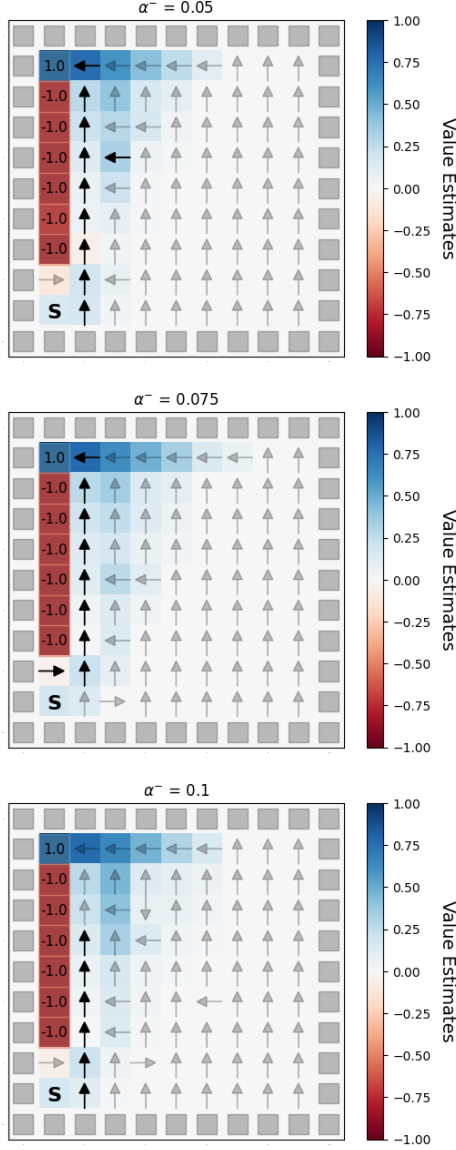


Figure 2: **Dyna α -SR agents' environment estimations.** The top image corresponds to the agent with $\alpha^- = 0.05$ (reward and punishment learning rates are equal), the middle to the agent with $\alpha^- = 0.075$ (50% higher than α^+), and the bottom to the agent with $\alpha^- = 0.1$ (double than α^+).

The three Dyna α -SR agents show a very similar behavior across the three analyses conducted. First, Figure 2 demonstrates that their environmental estimations are highly similar, even in the case where an agent had a punishment learning rate (α^-) that was twice the value of the reward learning rate (α^+). Additionally, all three agents chose a short but unsafe path to the goal during the testing phase (indicated by black arrows), underestimating the danger. While the first and second agents managed to reach the goal in the rollout, the third

agent followed a similar path but, due to the stochasticity in action selection, ended up falling into the cliff.

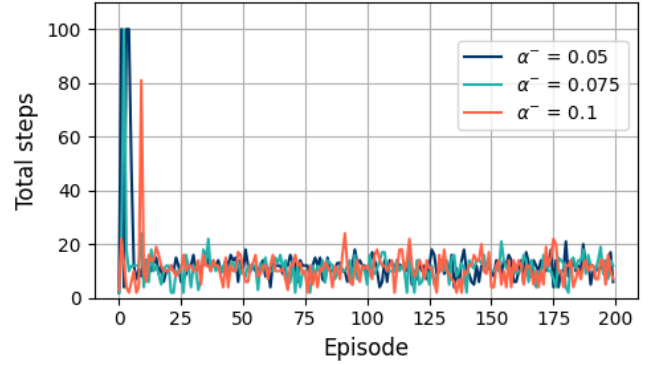


Figure 3: **Dyna α -SR agents' total steps per episode.**

Moreover, Figure 3 shows that, in general, the agents used a similarly small number of temporal steps across episodes, which is consistent with their preference for the shortest path to the goal. Only in a few episodes did the agents use more temporal steps than usual, probably because they ended up in an area that was still unexplored and farther from the cliff. Additionally, due to the similarities in both environmental estimations and the number of steps per episode, a similar level of goal achievement was expected. This is confirmed in Figure 4, which indicates that the agents mainly oscillated between reaching the goal (total return = 1) and falling into the cliff (total return = -1) across episodes, reflecting the inherent risk of choosing the shortest path. Finally, minor differences in behavior between the agents can be attributed to the stochasticity of the programmed policy.

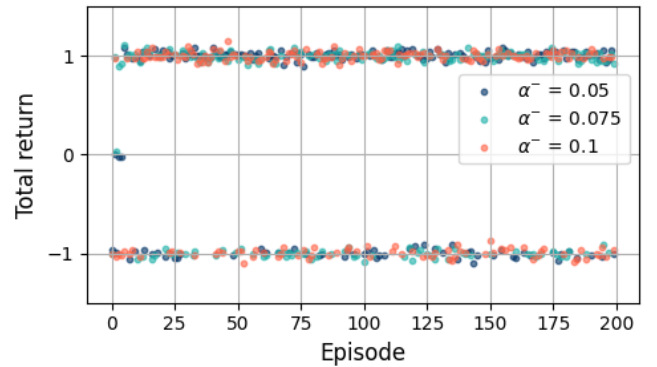


Figure 4: **Dyna α -SR agents' total return per episode.**

In summary, the results for the Dyna α -SR agents show that a higher punishment learning rate did not lead to the emergence of anxiety-related behaviors such as threat overestimation, fear generalization, avoidance, or risk aversion in the experimental task.

Dyna β -SR Agents

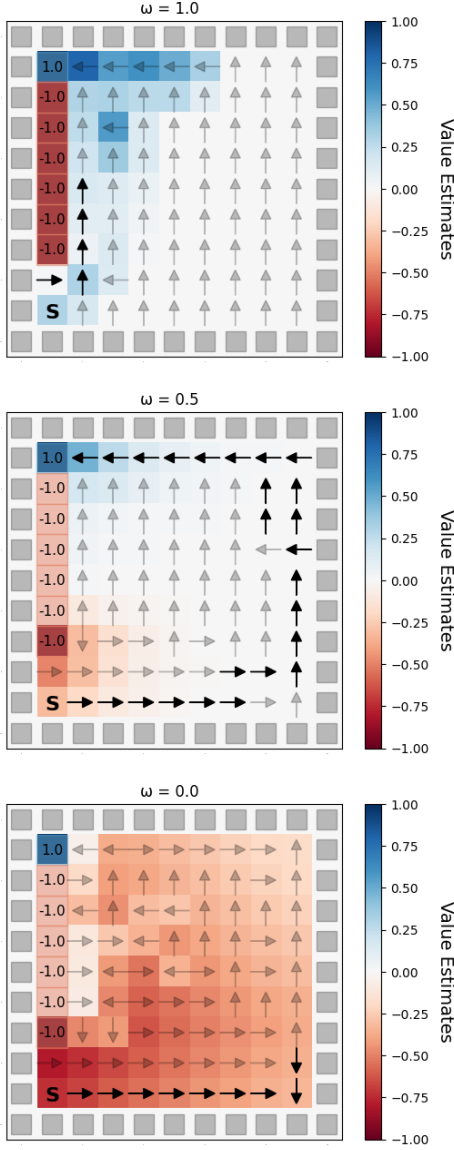


Figure 5: **Dyna β -SR agents' environment estimations.** The top image corresponds to the optimistic agent ($\omega = 1$), the middle to the agent with $\omega = 0.5$, and the bottom to the pessimistic agent ($\omega = 0$).

In contrast to Dyna α -SR agents, Dyna β -SR agents show a marked difference in learning and behavior across the analyses conducted. To begin with, Figure 5 shows that the optimistic agent ($\omega = 1$) learned to take a short but risky route to the goal near the cliff by estimating nearby states positively, without accounting for the possibility of not always being able to choose its own actions. Conversely, the moderately optimistic-pessimistic agent ($\omega = 0.5$) avoided the cliff during the testing phase, as it evaluated some nearby states as dangerous. Nevertheless, it was still able to reach the goal by

taking a longer but safer path. Finally, the pessimistic agent ($\omega = 0$) estimated much of the environment as aversive and tried to stay away from the cliff during the rollout, but ultimately failed to reach the goal.

Therefore, although the second and third agents exhibited avoidance behavior, it was the pessimistic one that excessively overestimated threat and demonstrated fear generalization. Additionally, the results suggest that threat overestimation and fear generalization bias reward evaluation, leading an agent to prioritize danger avoidance over goal pursuit. This reflects risk aversion, as shown by the third agent that exploited a safe zone of the state space instead of exploring the environment further.

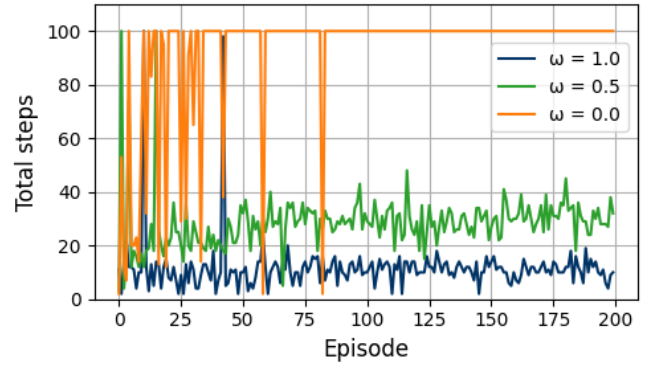


Figure 6: **Dyna β -SR agents' total steps per episode.**

Furthermore, the results from Figure 5 align with those in Figure 6, which show that the optimistic agent (blue) used the fewest temporal steps in most episodes by taking a more direct route to the goal. It was followed by the moderately optimistic-pessimistic agent (green), which took a longer but safer path to the goal. Meanwhile, the pessimistic agent (orange) used the maximum number of steps (100) in most episodes, as it tended to remain in neutral states far from danger.

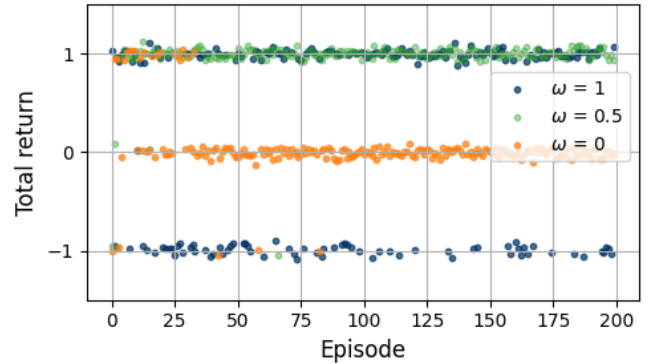


Figure 7: **Dyna β -SR agents' total return per episode.**

Lastly, the patterns observed for each agent in Figure 7 complement the previous results by highlighting how the

agents' environment estimations and learned strategies impacted their actual outcomes. The optimistic agent (blue) alternated between reaching the goal and falling into the cliff. The agent with $\omega = 0.5$ (green) achieved the goal most consistently, ending in neutral or aversive states only occasionally. In contrast, the pessimistic agent (orange) remained in neutral states for most of the episodes, particularly in the last 100.

In summary, the results for the Dyna β -SR agents indicate enhanced punishment sensitivity was directly linked to the exhibition of anxious behavior in the experimental task, as well as the necessity of a level of punishment sensitivity for an agent to identify a safer path to the goal.

Discussion

In this work, we expanded anxiety computational research in sequential evaluation tasks by using hybrid reinforcement learning models to assess the influence of punishment learning rate and punishment sensitivity in order to better understand the learning dynamics underlying anxious behavior. The results suggest that the impact of estimated punishments on planning is more significant than the speed at which they are learned, highlighting the importance of heightened punishment sensitivity in anxious decision-making.

Specifically, the behavioral similarity observed among Dyna α -SR agents indicates that a higher punishment learning rate does not promote the emergence of anxious behavior in the experimental task. In contrast, the performance differences among the Dyna β -SR agents support the idea that increased sensitivity to punishment facilitates the emergence of anxiety-related behaviors, such as avoidance, risk aversion, threat overestimation, and fear generalization. The Dyna β -SR model, in particular, predicts that pessimistic planning—arising from an enhanced sensitivity to threat due to the agent's underestimation of its own self-efficacy or coping resources—may play a central role in the development of these behaviors. This interpretation stems from the observation that what matters most is not how quickly punishments are learned, but how their estimations influence the updating of successor representations, which can lead to a progressive distortion of the cognitive map built by the agent when it has high sensitivity to punishments. In essence, the model highlights the role of catastrophic thinking or worry in anxiety, conceptualized here as a tendency to prioritize the most disadvantageous routes during both online and offline planning.

These findings align with the perspective presented by Brown et al. (2023), as well as with the sequential evaluation studies by Gagne and Dayan (2022); Zorowitz et al. (2020), which suggest that anxiety is primarily a disorder of uncertainty learning, where heightened punishment sensitivity might be a key feature in predicting anxious behavior. On the other hand, the results challenge aspects of prior research based on one-step tasks (Pike & Robinson, 2022), which have emphasized the role of rapid punishment integration in the emergence of anxious decision-making. This raises the ques-

tion of whether such tasks are adequate for studying anxiety, as they might not fully capture its complexity and could bias our understanding. Alternatively, it could be considered that different task structures—one-step versus multi-step—help clarify the specific ways in which a high punishment learning rate contributes to anxiety. For example, a high learning rate for punishments may be more relevant in the short term, as observed in one-step tasks, while it might lose relevance in sequential evaluation, where a high punishment sensitivity parameter becomes much more significant. Nevertheless, given that anxiety is primarily an anticipatory response to uncertain future threats perceived as uncontrollable and highly aversive (Clark & Beck, 2010), findings from sequential evaluation tasks may provide a more accurate understanding of the factors influencing anxious behavior.

Finally, this work provides valuable insights into the learning dynamics of anxiety considering sequential evaluation, although future research should explore other experimental tasks where differentiated learning rates may play a more significant role, particularly in non-deterministic environments. To achieve this, it will be essential to extend the models' offline planning module to account for uncertainty in environmental dynamics. Additionally, it will be interesting to determine the number of replays an agent makes, which can be linked to the degree of pessimism or surprise. Ultimately, an important next step should be to evaluate the Dyna β -SR model's predictions against human data to assess its plausibility beyond simulation.

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